SIMATS ENGINEERING

ASSESSMENT TOOL 3 – MODEL ANSWERS

Course: Computer Networks | Course Code: CSA07

Course Outcome CO2: Configure IP addressing schemes, subnetting, and basic routing techniques for efficient network design and topology.

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Q1. Debugging an IPv4 Address and Default Gateway

Given: IP address 192.168.10.65, subnet mask 255.255.255.192 (/26), default gateway 192.168.10.1.

1. Subnet calculation

- A /26 mask leaves 6 host bits. Each subnet contains $2^6 = 64$ total addresses, of which 64 − 2 = 62 are usable host addresses.
- The block size in the last octet is 256 − 192 = 64. The subnets in 192.168.10.0/24 therefore begin at .0, .64, .128 and .192.

Item	Correct result
Subnet containing 192.168.10.65	192.168.10.64/26
Network address	192.168.10.64
Broadcast address	192.168.10.127
Valid host range	192.168.10.65 – 192.168.10.126
Usable host addresses	62

2. Debugging findings

- The workstation IP 192.168.10.65 is valid in 192.168.10.64/26. It is not part of 192.168.10.0/26, whose range is network .0, usable hosts .1–.62, and broadcast .63.
- The configured gateway 192.168.10.1 belongs to 192.168.10.0/26, not to the workstation's /26 subnet. A host's directly connected default gateway should normally be in the same local subnet.
- The stated failure is therefore explained by a subnet/gateway mismatch if the intended network is 192.168.10.0/26. The IP and mask themselves are valid together, but for a different subnet.

3. Corrected configuration

Setting	Recommended value
IP address	192.168.10.10
Subnet mask	255.255.255.192
Default gateway	192.168.10.1
Network	192.168.10.0/26
Usable hosts after correction	62 (192.168.10.1–192.168.10.62; reserve .1 for gateway)

Verification: confirm the router interface is configured as 192.168.10.1/26; check the PC's IP, mask and gateway; then ping 192.168.10.1 and another host in the subnet. In Packet Tracer, inspect the PC IP Configuration and router interface status.

Q2. Debugging IPv6 Addressing

Given: Device A = 2001:db8::ff00:42:8329/64; Device B = 2001:db8:0:0:1::1/64.

1. Expand the addresses

Device	Full eight-hextet notation	/64 network prefix
A	2001:0db8:0000:0000:0000:ff00:0042:8329	2001:0db8:0000:0000::/64
B	2001:0db8:0000:0000:0001:0000:0000:0001	2001:0db8:0000:0000::/64

Important: a /64 prefix is determined by the first four hextets. For both addresses as written, those first four hextets are 2001:0db8:0000:0000. Therefore, both addresses are in the same /64 subnet. The apparent extra "1" in Device B is in the interface identifier (the fifth hextet), not in the /64 network prefix.

2. Debugging findings and correction

- The two given addresses do not have a /64 prefix mismatch. Both belong to 2001:db8:0:0::/64.
- If communication fails, investigate other causes: incorrect interface prefix length, duplicate address, disabled interface, VLAN or link issue, missing/incorrect IPv6 routing when crossing networks, ICMPv6 filtering, or a wrong default gateway for off-link destinations.
- Example valid same-subnet configuration: Device A 2001:db8:0:0::10/64 and Device B 2001:db8:0:0::20/64. These are examples only; do not assign them if already in use.

3. Available addresses

- A /64 leaves 64 bits for the interface identifier: 2^{64} = 18,446,744,073,709,551,616 possible interface-ID values.
- IPv6 does not use the IPv4-style network/broadcast reservation rule; there is no broadcast address. In practice, usable assignment is subject to IPv6 conventions and reserved values.

Verification: use ipconfig /all or the operating system's IPv6 interface display, verify the prefix length is /64, and test pinging the peer's IPv6 address. In Wireshark, inspect ICMPv6 Neighbor Solicitation/Advertisement and Echo Request/Reply packets.

Q3. Analyze and Optimize IPv4 Subnet Allocation Using VLSM

Given: 192.168.1.0/24 is divided using /26, /27 and /28 masks. The task does not specify the number of hosts required by each department, so an exact utilization calculation cannot be finalized without those requirements. The following illustrates the address ranges and a sample VLSM plan.

1. Subnet capacities

Prefix / mask	Total addresses	Usable IPv4 hosts	Block size
/26 – 255.255.255.192	64	62	64
/27 – 255.255.255.224	32	30	32
/28 – 255.255.255.240	16	14	16

2. Example valid non-overlapping allocation

Subnet	Network address	Usable host range	Broadcast	Usable hosts
Department A – /26	192.168.1.0	192.168.1.1 – 192.168.1.62	192.168.1.63	62
Department B – /27	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95	30
Department C – /28	192.168.1.96	192.168.1.97 – 192.168.1.110	192.168.1.111	14

This is one valid allocation example. It uses 112 addresses in three subnet blocks (64 + 32 + 16), leaving 144 addresses from 192.168.1.112 through 192.168.1.255 unallocated. These remaining addresses are not automatically "wasted"; they may be reserved for future growth or further subnets.

3. Utilization and common allocation errors

- To calculate unused usable host capacity for each department, the actual required host count is needed. Formula: unused usable capacity = usable host count − required hosts.
- A /26 provides 62 usable addresses, a /27 provides 30, and a /28 provides 14. Assigning /26 to a department needing only a few addresses may waste space; assigning /28 to a department needing more than 14 hosts is insufficient.
- Common errors include overlapping subnet ranges, assigning a network or broadcast address to a host, using a gateway outside the subnet, and choosing a prefix that cannot accommodate the required hosts.

4. Recommended VLSM method

1. List each department's current host requirement and expected growth.
2. Sort departments from largest requirement to smallest.
3. For each department, choose the smallest subnet where $2^h - 2$ is at least the required host count (for ordinary IPv4 subnets).
4. Allocate each subnet on its correct block-size boundary, without overlap; reserve room for growth and infrastructure.
5. Document the network, mask, usable range, broadcast, gateway and reserved addresses.

Illustrative example only: if departments require 50, 25 and 12 hosts, allocate /26 (62 usable), /27 (30 usable) and /28 (14 usable), respectively. This is efficient for those assumed requirements. Replace these assumed counts with the organization's actual host requirements before presenting the final plan.

Q4. IPv6 Debugging (Duplicate of Q2 in the Assessment)

Q4 repeats the same IPv6 scenario as Q2. The correct analysis is therefore the same: expand the addresses, compare the first four hextets for a /64, and troubleshoot beyond prefix mismatch because both supplied addresses belong to 2001:db8:0:0::/64.

Device	Full address	Prefix
A	2001:0db8:0000:0000:0000:ff00:0042:8329/64	2001:db8:0:0::/64
B	2001:0db8:0000:0000:0001:0000:0000:0001/64	2001:db8:0:0::/64

Suggested action: verify both interfaces use /64, confirm the devices share the same Layer 2/VLAN segment when expected, check duplicate-address detection and Neighbor Discovery, and inspect ICMPv6 traffic in Wireshark. A /64 has 2^{64} possible interface identifier values; IPv6 has no broadcast address.

Q5. Debugging a Routing Table with Longest-Prefix Match

Given routing entries:

Destination network	Next hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default route (0.0.0.0/0)	10.0.0.1

1. Route selection for 192.168.2.100

- 192.168.2.100 falls within 192.168.2.0/24 because the /24 range is 192.168.2.0–192.168.2.255.
- The longest matching prefix is 192.168.2.0/24, so the selected next hop is 10.0.0.3.
- The default route 0.0.0.0/0 is less specific and is used only when no more-specific route matches.

2. Debugging and optimization

- The shown table selects the expected route for 192.168.2.100. If packets still fail, the fault may be beyond this table: next hop 10.0.0.3 may be unreachable, the next-hop router may lack a return route, an interface may be down, an ACL/firewall may block traffic, or the destination host may be misconfigured.
- Verify that 10.0.0.3 is reachable on a directly connected or otherwise reachable network, confirm the remote router has a route back to the source network, and inspect interface status and packet flow.
- The table is already logically optimized for the stated destination because it contains a specific /24 route and a default route. Avoid adding redundant or overlapping routes unless required by the topology.

Destination tested	Matching route	Next hop used
192.168.2.100	192.168.2.0/24	10.0.0.3
192.168.1.25	192.168.1.0/24	10.0.0.2
203.0.113.10 (example with no specific match)	0.0.0.0/0 default route	10.0.0.1

The example address 203.0.113.10 is from a documentation-only range and is used here only to demonstrate default-route selection.

Practical Verification Checklist

- In Cisco Packet Tracer: check host IP address, subnet mask, default gateway, router interface IPs and interface status.
- Use ping to test local gateway reachability, peer-to-peer reachability and remote-network reachability in sequence.
- Use tracert/traceroute where supported to identify the hop at which forwarding fails.
- In Wireshark, inspect ARP for IPv4 or ICMPv6 Neighbor Discovery for IPv6, then inspect ICMP Echo Request/Reply and relevant routing/ICMP errors.
- Record the actual test result and include screenshots from your own Packet Tracer/Wireshark setup if required by your instructor.

Conclusion

The key debugging principle is to validate addressing first, then the local gateway and Layer 2 connectivity, and finally routing and return paths. For Q1, the IP belongs to 192.168.10.64/26 while the given gateway belongs to 192.168.10.0/26; changing the workstation to an unused address such as 192.168.10.10/26 aligns it with the intended subnet. For Q2 and Q4, both IPv6 addresses share the same /64 prefix, so the stated prefix-mismatch diagnosis is not supported by the supplied addresses. For Q3, VLSM should be based on real host counts. For Q5, longest-prefix match selects 192.168.2.0/24 via 10.0.0.3.